Question 1A [10 Marks]

In round one of a scavenger hunt, organizers have placed vital information in a locked metal box. The password to unlock the box is its area. The only information that is available about the box is the curves that bound the surface of the box; $e^y = x$; $y - 2 = 0$; $-x + y + 1 = 0$. Those participants who will successfully calculate the correct area of the box will be able to unlock the box and get the clue that is hidden inside the box. Sketch the graph and find the area under the curve.

$e^y = x$, $y - 2 = 0$, $-x + y + 1 = 0$

For 1st area problem we need to convert ftns into $y = f(x)$ form

$e^y = x$

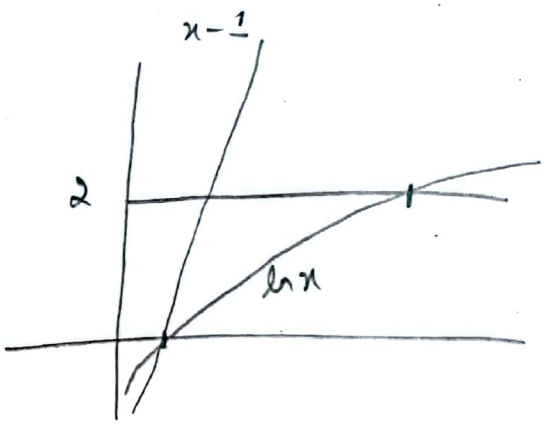
$\ln(e^y) = \ln x$

$y \ln e = \ln x$

$y = \ln x \longrightarrow$ (a)

$y = 2 \longrightarrow$ (b)

$y = x - 1 \longrightarrow$ (c)



For finding pts of intersection equate (a) & (b), (a) & (c), (b) & (c)

$\ln x = 2$

$\boxed{x = e^2}$

$x - 1 = 2$

$\boxed{x = 3}$

$\ln x = x - 1$

$e^{\ln x} = e^{x-1}$

$x = e^{x-1}$

$\boxed{x = 1}$

$$\text{Area} = \int_a^b [\text{upper fn} - \text{lower fn}]\, dx + \int_c^d [\text{upper fn} - \text{lower fn}]\, dx$$

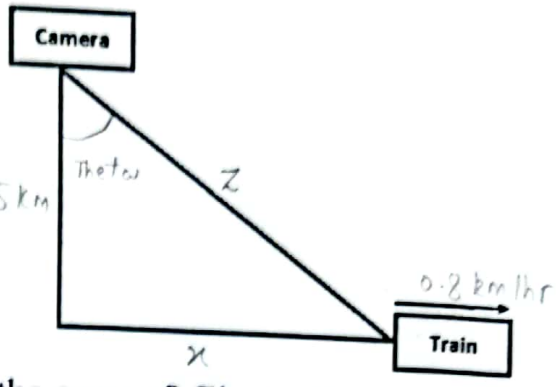
$$\text{Area} = \int_1^3 [x - 1 - \ln x]\, dx + \int_3^{e^2} [2 - \ln x]\, dx$$

$$\text{Area} = \left[\frac{x^2}{2} \Big|_1^3 - x \Big|_1^3 - \frac{1}{x} \Big|_1^3 \right] + \left[2x \Big|_3^{e^2} - \frac{1}{x} \Big|_3^{e^2} \right]$$

$$\text{Area} = e^2 - 5$$

Question 1B [10 Marks]

A train is traveling at 0.8 km/min along a straight track, moving in the direction as shown below. A movie camera, 0.5 km away from the track is focused on the train.



a) Express z, the distance between the camera and the train as function of x.

b) How fast is the distance from the camera to the train changing when the train is 1km from the camera? Give units.

c) How fast is the camera rotating (in radians/min) at the moment when the train is 1 km from the camera?

(a) Using Pythagoras Theorem, we have $x^2 + (0.5)^2 = z^2$

so

$$z = \sqrt{x^2 + \frac{1}{4}}$$

(b) z : The distance from the Camera to the train

x : Distance from where the train directly passes the Camera

t : Time

$\frac{dz}{dt}$: Rate of change of distance from the Camera to the train

$\frac{dx}{dt} = 0.8$ (distance x is increasing at a rate of 0.8 km/hr)

$$x^2 + \frac{1}{4} = z^2 \quad \rightarrow (i)$$

Differentiating (i) w.r.t t

$$2x \frac{dx}{dt} = 2z \frac{dz}{dt}$$

$$\frac{dz}{dt} = \frac{x}{z} \frac{dx}{dt}$$

When $z = 1$, $x^2 = z^2 - (0.5)^2$

$$x^2 = 1 - 0.25$$

$$x^2 = 0.75$$

$$x = \sqrt{3}/2$$

$$\frac{dz}{dt} = \frac{(\sqrt{3}/2)}{1}(0.8)$$

$$\frac{dz}{dt} = \frac{\sqrt{3}}{2}(0.8) \text{ km/hr}$$

(c) $\frac{d\theta}{dt} = ?$

Using trigonometry

$$\tan\theta = \frac{x}{0.5}$$

$$\tan\theta = 2x \rightarrow (ii)$$

$$\theta = \tan^{-1}(2)\left(\frac{\sqrt{3}}{2}\right)$$

$$\theta = \tan^{-1}(\sqrt{3})$$

$$\theta = \frac{\pi}{3}$$

Differentiating (ii) w.r.t t

$$\sec^2\theta \frac{d\theta}{dt} = 2\frac{dx}{dt}$$

$$\frac{1}{\cos^2\theta}\frac{d\theta}{dt} = 2\frac{dx}{dt}$$

$$\frac{d\theta}{dt} = 2\cos^2\theta \frac{dx}{dt}$$

$\therefore \cos\left(\frac{\pi}{3}\right) = 0.5$

$$\frac{d\theta}{dt} = 2(0.5)^2(0.8)$$

$$\frac{d\theta}{dt} = 0.4 \text{ rad/min}$$

Question 2A [10 Marks]

Use the integral test to determine if the following series converge or diverge. Be sure to check that the conditions of the integral test are satisfied.

$$\sum_{k=1}^{\infty} \frac{k^2}{e^{\frac{k}{3}}}$$

* $f(x) = \frac{x^2}{e^{x/3}}$ is +ive & Continuous for $x \geq 1$

* $f'(x) = \frac{6x - x^2}{3e^{x/3}} = \frac{-x(x-6)}{3e^{x/3}} < 0$ when $x > 6$

$f'(x)$ is decreasing when $x > 6$.

* $\int_7^{\infty} f(x)\,dx = \int_7^{\infty} \frac{x^2}{e^{x/3}}\,dx \Rightarrow \int_7^{\infty} x^2 e^{-x/3}\,dx$

Let $u = x^2$, $dv = e^{-x/3}dx$, $du = 2x\,dx$, $v = -3e^{-x/3}$

$$x^2(-3e^{-x/3})\Big|_7^{\infty} - \int_7^{\infty} 2x(-3e^{-x/3})\,dx$$

$$\frac{-3x^2}{e^{x/3}}\Big|_7^{\infty} - \int_7^{\infty} 2x(-3e^{-x/3})\,dx$$

Let $u = 2x$, $dv = -3e^{-x/3}$, $du = 2dx$, $v = 9e^{-x/3}$

$$\frac{-3x^2}{e^{x/3}}\Big|_7^{\infty} - \left[(2x)(9e^{-x/3})\Big|_7^{\infty} - \int_7^{\infty} 9e^{-x/3} \cdot 2dx\right]$$

$$\left[\frac{-3x^2}{e^{x/3}} - \frac{18x}{e^{x/3}} - \frac{54}{e^{x/3}}\right]_7^{\infty}$$

$$\lim_{b\to\infty}\left[-\frac{3b^2}{e^{b/3}}-\frac{18b}{e^{b/3}}-\frac{54}{e^{b/3}}\right]+\frac{3(7^2)+18(7)+54}{e^{7/3}}$$

$$=\frac{327}{e^{7/3}}$$

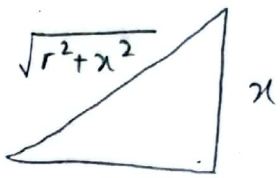
Improper Integral converges so is the series.

Question 2B [10 Marks]

In electromagnetic theory, the magnetic potential at a point on the axis of circular coil is given by

$$U = \frac{2\pi NIr}{k}\int_{a}^{+\infty} \frac{dx}{(r^2+x^2)^{\frac{3}{2}}}$$

where N, I, r, k and a are constants. Find U.



$x = r\tan\theta$

Let $x = r\tan\theta$

$dx = r\sec^2\theta\, d\theta$

$$\int \frac{dx}{(\sqrt{r^2+x^2})^3} = \int \frac{r\sec^2\theta\, d\theta}{(\sqrt{r^2+r^2\tan^2\theta})^3}$$

$$= \int \frac{r\sec^2\theta\, d\theta}{(r\sqrt{1+\tan^2\theta})^3}$$

$$= \int \frac{r\sec^2\theta\, d\theta}{r^3(\sqrt{\sec^2\theta})^3}$$

$$= \int \frac{\sec^2\theta\, d\theta}{r^2\sec^3\theta}$$

$$= \int \frac{d\theta}{r^2\sec\theta}$$

$$= \frac{1}{r^2}\int \frac{1}{1/\cos\theta}\, d\theta$$

$$= \frac{1}{r^2}\int \cos\theta\, d\theta$$

$$= \frac{1}{r^2}\sin\theta + C$$

$$= \frac{1}{r^2}\cdot\frac{x}{\sqrt{r^2+x^2}} + C$$

$$U = \frac{2\pi N I r}{k} \cdot \frac{1}{r^2} \cdot \frac{x}{\sqrt{r^2+x^2}}$$

$$U = \frac{2\pi N I}{rk} \cdot \frac{x}{\sqrt{r^2+x^2}}$$

$$U = \frac{2\pi N I}{rk} \lim_{b\to\infty} \left(\frac{x}{\sqrt{r^2+x^2}} \Big|_a^b \right)$$

$$U = \frac{2\pi N I}{rk} \lim_{b\to\infty} \left[\frac{b}{\sqrt{r^2+b^2}} - \frac{a}{\sqrt{r^2+a^2}} \right]$$

$$U = \frac{2\pi N I}{rk} \left[1 - \frac{a}{\sqrt{r^2+a^2}} \right]$$

Ag $\lim_{b\to\infty} \frac{b}{\sqrt{r^2+b^2}}$

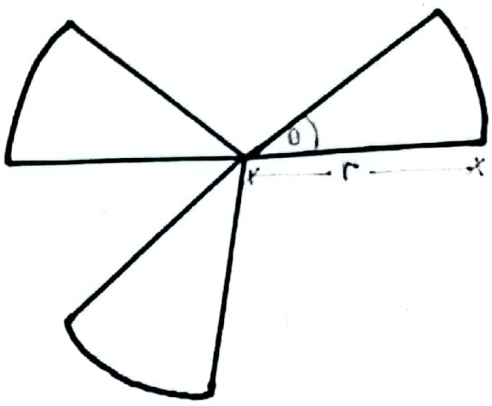
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Constant

$$\lim_{b\to\infty} \frac{b}{\sqrt{b^2}} = 1$$

Question 3 [10 Marks]

The figure shows the design of a hazard warning logo which consists of three identical sectors of radius r cm, joined together at the center. Each sector subtends an angle θ radians at the center and the sectors are equally spaced so that the logo has rotational symmetry of order 3. The area of the logo is $75cm^2$.



a) Find out the perimeter, P cm, of the logo.

b) Determine the value of r for which P is stationary.

c) Show that the value of r found in part (b) gives the minimum value for P.

d) Find the minimum perimeter.

(a) $P = 6r + 3r\theta \longrightarrow (i)$

Constraint $A = 75$

$\left(\frac{1}{2} r^2\theta\right) \cdot 3 = 75$

$\frac{1}{2} r^2\theta = 25$

$r^2\theta = 50$

$r\theta = \frac{50}{r}$



arc length $= S = r\theta$

Substituting value of '$r\theta$' in (i) we get

$P = 6r + 3\left(\frac{50}{r}\right)$

$P = 6r + \frac{150}{r}$

(b) $P = 6r + 150r^{-1}$

$\frac{dP}{dr} = 6 - 150r^{-2}$

$\frac{dP}{dr} = 0$

$$6 - \frac{150}{r^2} = 0$$

$$r^2 = 25$$

$$r = 5 \qquad (r > 0)$$

(c) $$\frac{d^2P}{dr^2} = 300r^{-3}$$

$$\frac{d^2P}{dr^2} = \frac{300}{r^3}$$

$$\left.\frac{d^2P}{dr^2}\right|_{r=5} = \frac{300}{125} = 2.4 > 0$$

Indeed a minimum.

(d) $$P = 6(5) + \frac{150}{5}$$

$$P_{min} = 60 \text{ cm}$$

Q5

$$\int_{\ln(\frac{3}{4})}^{\ln(\frac{4}{3})} \frac{e^t}{(1+e^{2t})^{3/2}}\,dt$$

Let $e^t = \tan\theta$

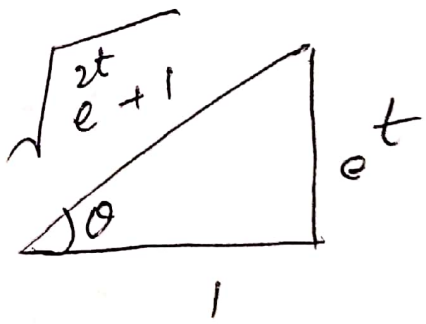
$$e^t dt = \sec^2\theta\, d\theta$$

$$I = \int \frac{\sec^2\theta\, d\theta}{(1+\tan^2\theta)^{3/2}} = \int \frac{\sec^2\theta\, d\theta}{(\sec^2\theta)^{3/2}}$$

$$I = \int \frac{\sec^2\theta\, d\theta}{\sec^3\theta} = \int \frac{1}{\sec\theta}\, d\theta$$

$$I = \int \cos\theta\, d\theta = \sin\theta + C$$

$$\left(\sec^2\theta = 1 + e^{2t} \qquad \sec\theta = \pm\sqrt{1+e^{2t}}\right)$$



$$I = \frac{e^t}{\sqrt{e^{2t}+1}}\Bigg|_{\ln(3/4)}^{\ln(4/3)}$$

$$I = \frac{e^t}{\sqrt{e^{2t}+1}}\Bigg|_{\ln(3/4)}^{\ln(\frac{4}{3})}$$

$$I = \frac{\frac{4}{3}}{\sqrt{(\frac{4}{3})^2+1}} - \frac{\frac{3}{4}}{\sqrt{(\frac{3}{4})^2+1}} = \frac{4}{\sqrt{16+9}} - \frac{3}{\sqrt{9+16}}$$

$$I = \frac{4}{5} - \frac{3}{5} = \frac{1}{5} \text{ kg}$$

(C)

Q4a

$$f(x) = x - \left(\frac{1}{2}\right)^x, \quad F(1) = -1$$

The Anti-derivative of $f(x)$ is $F(x)$

$$F(x) = \int f(x)\,dx$$

$$F(x) = \int x\,dx - \int \left(\frac{1}{2}\right)^x dx$$

$$F(x) = \frac{x^2}{2} - \frac{\left(\frac{1}{2}\right)^x}{\ln\left(\frac{1}{2}\right)} + C$$

$$F(x) = \frac{x^2}{2} - \frac{\left(\frac{1}{2}\right)^x}{\ln(1) - \ln 2} + C$$

$$F(x) = \frac{x^2}{2} + \frac{\left(\frac{1}{2}\right)^x}{\ln 2} + C \quad \text{— (A)}$$

$$F(1) = -1$$

$$-1 = \frac{1}{2} + \frac{\frac{1}{2}}{\ln 2} + C$$

$$\frac{-3}{2} = \frac{1}{2\ln 2} + C$$

$$C = -3 - \frac{1}{\ln 2}$$

Put in $C = -3 - \frac{1}{\ln 2}$ in (A)

(3

$$F(x) = \frac{x^2}{2} + \frac{(2)^{-x}}{\ln 2} - 3 - \frac{1}{\ln 2}$$

$$F(x) = \left(\frac{x^2}{2} - 3\right) + \frac{1}{\ln 2}\left[(2)^{-x} - 1\right]$$

Q4b

write domain and Range

x = hours

y = cost

$x = 0 \rightarrow y = 80$

$x = 8 \rightarrow y = 440$

Domain; $0 \leq x \leq 8$

Range, $80 \leq x \leq 440$

Q6a prove that

$$\lim_{x\to 0} x^4 \cos\left(\frac{2e^2}{x}\right) = 0$$

Sol:-

Since

$$-1 \le \cos\left(\frac{2e^2}{x}\right) \le 1$$

$$-x^4 \le x^4 \cos\left(\frac{2e^2}{x}\right) \le x^4$$

Apply $\lim_{x\to 0}$

$$\lim_{x\to 0}(-x^4) \le \lim_{x\to 0} x^4 \cos\left(\frac{2e^2}{x}\right) \le \lim_{x\to 0} x^4$$

Since

$$\lim_{x\to 0}(-x^4) = \lim_{x\to 0} x^4 = 0$$

By Squeeze theorem.

$$\lim_{x\to 0} x^4 \cos\left(\frac{2e^2}{x}\right) = 0$$

Q6b Sol:-

Given $I = \int_0^{\pi/4} \tan^6 x \sec x \, dx$

Let $J = \int_0^{\pi/4} \tan^8 x \sec x \, dx$

$J = \int_0^{\pi/4} \tan^7 x \cdot \sec x \tan x \, dx$

u $\quad$ dv

$u = \tan^7 x \qquad \int dv = \int \sec x \tan x \, dx$

$du = 7 \tan^6 x \cdot \sec^2 x \, dx \qquad v = \sec x$

$J = uv - \int v \, du$

$= \tan^7 x \sec x \Big|_0^{\pi/4} - \int_0^{\pi/4} 7 \tan^6 x \cdot \sec^3 x \, dx$

$= \sqrt{2} - 7 \int_0^{\pi/4} \tan^6 x \cdot \sec x (1 + \tan^2 x) \, dx$

$= \sqrt{2} - 7 \int_0^{\pi/4} \tan^6 x \sec x - 7 \int_0^{\pi/4} \tan^8 x \sec x \, dx$

$J = \sqrt{2} - 7I - 7J$

$J = \frac{\sqrt{2}}{8} - \frac{7}{8} I$

Q7

Sol:-

$$\sum_{n=1}^{\infty} \frac{(-1)^n 3^n + 5^{-n}}{15^n}$$

$$= \sum_{n=1}^{\infty} \left(\frac{(-1)^n 3^n}{15^n} + \frac{5^{-n}}{15^n} \right)$$

$$= \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{5^n} + \frac{1}{3^n} \right)$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n}{5^n} + \sum_{n=1}^{\infty} \frac{1}{3^n}$$

$\sum_{n=1}^{\infty} ar^n$

$|r| < 1 \Rightarrow$ Converges

$\sum_{n=1}^{\infty} \frac{(-1)^n}{5^n}$ is a Geometric series with $|r| = \left|\frac{-1}{5}\right| < 1$

$\Rightarrow$ Converges.

$\sum_{n=1}^{\infty} \frac{1}{3^n}$ is a Geometric series

with $|r| = \frac{1}{3} < 1 \Rightarrow$ Converges.

Hence the given series converges.

Question # 8(a):

Solve the following integral

$$\int \frac{\mathbf{1}}{(\sqrt{x})\left(e^{-\sqrt{x}}\right)} Sec^2\left(e^{\sqrt{x}+1}\right)\, dx$$

Solution:

Let, $y = \sqrt{x} + 1 \quad \Rightarrow y - 1 = \sqrt{x}$

$\frac{dy}{dx} = \frac{1}{2(\sqrt{x})} \quad \Rightarrow 2dy = \frac{1}{\sqrt{x}}$

$$\int \frac{Sec^2(e^y)}{e^{1-y}}(2dy) = \frac{2}{e}\int e^y Sec^2(e^y)dy$$

$= \frac{2}{e}\int (Sec^2 v)dv \quad also\ substitute \quad v = e^y \quad , \frac{dv}{dy} = e^y \quad , \quad dv = e^y dy$

$= \frac{2}{e}\tan|v| + c$

$= \frac{2}{e}\tan|e^y| + c$

$= \frac{2}{e}\tan\left|e^{\sqrt{x}+1}\right| + c$

Question #8(b).

Test the convergence or divergence of the series by using the suitable test.

$$\sum_{n=1}^{\infty} \frac{\mathbf{3^n}}{\mathbf{n^3 2^n}}$$

Solution:

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{3^{n+1}}{(n+1)^3 2^{n+1}} \times \frac{2^n n^3}{3^n}\right|$$

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{3}{2}\left(\frac{n}{n+1}\right)^3$$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\frac{3}{2}\left(\frac{1}{1+\left(\frac{1}{n}\right)}\right)^3$$

$= \frac{3}{2} < 1 \quad thus\ the\ given\ series\ is\ divergent$

9a

Sketch the graph

$$f(x) = x^4 - 6x^2 + 5$$

Sol:-

$$f'(x) = 4x^3 - 12x$$

$$\Rightarrow f'(x) = 0 \quad , \quad x = 0, \sqrt{3}, -\sqrt{3}$$

Interval	$x < -\sqrt{3}$	$-\sqrt{3} < x < 0$	$0 < x < \sqrt{3}$	$x > \sqrt{3}$
Sign of f'	-ve	+ve	-ve	-ve
Behavior of f	decreasing	Increasing	decreasing	decreasing.

By using first derivative test, the function has local minimum at $x = -\sqrt{3}$ and local maximum at $x = 0$

$$f''(x) = 12x^2 - 12$$

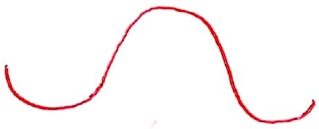
$$f''(x) = 0 \Rightarrow x = \pm 1$$

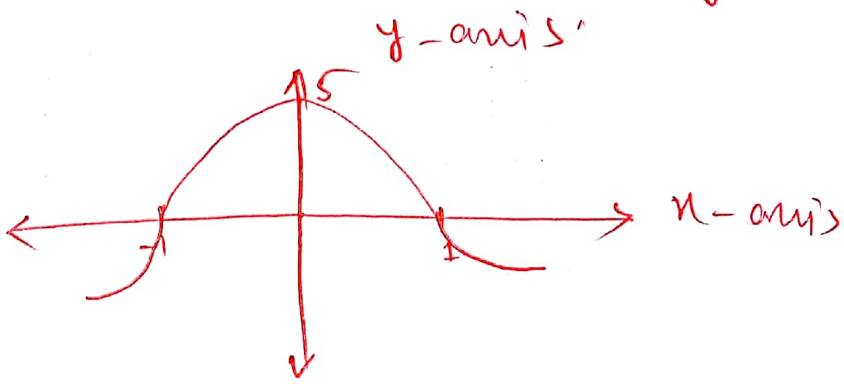
Interval :-	$x < -1$	$-1 < x < 1$	$x > 1$
Sign of f''	+ ve	- ve	+ ve
Behavior of f	Concave up	Concave down	Concave up.

$x < -\sqrt{3}$	$-\sqrt{3} < x < -1$	$-1 < x < 0$	$0 < x < 1$	$1 < x < \sqrt{3}$	$x > \sqrt{3}$
decreasing	Increasing	Increasing	decreasing	decreasing	decreasing
Concave up	Concave up	Concave down	Concave down	concave up	Concave up

So, $x = -1$ and $x = +1$ are the two points of inflection.

The general shape of the graph of the function is given by.





Q9b $\int \cos 2x \sin 3x \, dx$

let

$$I = \int \cos 2x \sin 3x \, dx$$

$$= \frac{1}{2}\sin 2x \sin 3x - \frac{3}{2}\int \sin 2x \cos 3x \, dx$$

$$= \frac{1}{2}\sin 2x \sin 3x - \frac{3}{2}\left[-\frac{1}{2}\cos 3x \cos 2x - \frac{3}{2}\int \sin 3x \cos 2x \, dx\right]$$

$$= \frac{1}{2}\sin 2x \sin 3x + \frac{3}{4}\cos 3x \cos 2x + \frac{9}{4}\int \sin 3x \cos 2x \, dx$$

$$I = \frac{1}{2}\sin 2x \sin 3x + \frac{3}{4}\cos 3x \cos 2x + \frac{9}{4} I$$

$$\Rightarrow \left(1 - \frac{9}{4}\right) I = \frac{1}{4}\left[2 \sin 2x \sin 3x + 3 \cos 3x \cos 2x\right]$$

$$\Rightarrow -\frac{5}{4} I = \frac{1}{4}\left[2 \sin 2x \sin 3x + 3 \cos 3x \cos 2x\right]$$

$$\Rightarrow I = -\frac{1}{5}\left[2 \sin 2x \sin 3x + 3 \cos 3x \cos 2x\right]$$

$$\Rightarrow \int \cos 2x \sin 3x \, dx = -\frac{2}{5}\sin 2x \sin 3x - \frac{3}{5}\cos 3x \cos 2x$$

Ans